The Method of Least Squares: Methods

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Outline I

Outline II

Hierarchy of Least Squares Methods

1. normal equations,
2. **QR** decomposition,
3. singular value decomposition,
4. iterative refinement

Linear System

Shape of linear system for Bevington data:

$$
\begin{array}{cccc}
\mathbf{A} & a & = & T \\
\begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} & \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} & = & \begin{bmatrix} T_1 \\ \vdots \\ T_m \end{bmatrix}
\end{array}
\tag{1}
$$

Linear System with Mesh and Data Vectors

$$
\begin{array}{cccc}
\mathbf{A} & a & = & T \\
\begin{bmatrix} \mathbf{1} & x \end{bmatrix} & \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} & = & \begin{bmatrix} T \end{bmatrix} \\
\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix} & \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} & = & \begin{bmatrix} 15.6 \\ 17.5 \\ 36.6 \\ 43.8 \\ 58.2 \\ 61.6 \\ 64.2 \\ 70.4 \\ 98.8 \end{bmatrix}
\end{array}
\tag{2}
$$

Form the Normal Equations

$$
\begin{aligned}
\mathbf{A}^*\mathbf{A} \quad & a & = \quad & \mathbf{A}^* T \\
\begin{bmatrix} \mathbf{1}^{\mathrm{T}}\mathbf{1} & \mathbf{1}^{\mathrm{T}}x \\ x^*\mathbf{1} & x^*x \end{bmatrix} & \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} & = \quad & \begin{bmatrix} \mathbf{1}^{\mathrm{T}}T \\ x^*T \end{bmatrix} \\
\begin{bmatrix} 9 & 45 \\ 45 & 285 \end{bmatrix} & \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} & = \quad & \begin{bmatrix} 466.7 \\ 2898 \end{bmatrix}
\end{aligned} \tag{3}
$$

Solve the Normal Equations

$$
\begin{aligned}
a_p &= \left(\mathbf{A}^*\mathbf{A}\right)^{-1} \mathbf{A}^* T, \\
\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p &= \Delta^{-1} \begin{bmatrix} x^* x & -\mathbf{1}^{\mathrm{T}} x \\ -x^* \mathbf{1} & \mathbf{1}^{\mathrm{T}} \mathbf{1} \end{bmatrix} \begin{bmatrix} \mathbf{1}^{\mathrm{T}} T \\ x^* T \end{bmatrix}, \\
& \frac{1}{540} \begin{bmatrix} 285 & -45 \\ -45 & 9 \end{bmatrix} \begin{bmatrix} 466.7 \\ 2898 \end{bmatrix}.
\end{aligned}
\tag{4}
$$

Solution as Rational Numbers

$$a_{LS} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \frac{1}{360} \begin{bmatrix} 1733 \\ 3387 \end{bmatrix} \approx \begin{bmatrix} 4.8 \\ 9.81 \end{bmatrix}. \tag{5}$$

Error Propagation

Least total squared error:

$$t^2 = r^*(a_p) r(a_p) = \frac{1\,139\,969}{3600} \approx 317. \tag{6}$$

Uncertainty in fit parameters:

$$\sigma_k = \sqrt{\frac{t^2}{m-n} \left(\mathbf{A}^* \mathbf{A}\right)^{-1}_{kk}} \approx \begin{bmatrix} 4.9 \\ 0.87 \end{bmatrix}. \tag{7}$$

Orthogonalize $\mathcal{R}(\mathbf{A})$

The first vector only needs to be normalized:

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \mathbf{1}_9 \Rightarrow v_1 = \frac{u_1}{\|u_1\|_2} = \frac{1}{3}\mathbf{1}_9 \tag{8}$$

Second Column Vector

The second column vector contains the mesh

$$u_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix} \tag{9}$$

Second Column Vector

Remove the projection of v_1 onto u_2. The projection operator is

$$P_{u_2}\left(v_1\right) = \frac{v_1^{\mathrm{T}} u_2}{u_2^{\mathrm{T}} u_2} \tag{10}$$

The purified second column vector is

$$\hat{v}_2 = u_2 - P_{u_2}(v_1) \tag{11}$$

Normalize the second column vector $v_2 = \frac{\hat{v}_2}{\|\hat{v}_2\|_2}$

Assemble Matrix Q

The matrix is

$$\mathbf{Q} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} & \frac{1}{2\sqrt{15}} \begin{bmatrix} -4 \\ -3 \\ -2 \\ -1 \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \end{bmatrix} \tag{12}$$

Orthogonality

$$\mathbf{Q}^{*}\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{13}$$

R Matrix

The upper triangular $\mathbf{R}$ matrix collects the blue factors:

$$\mathbf{R} = \begin{bmatrix} 3 & 15 \\ 0 & 2\sqrt{15} \end{bmatrix} \tag{14}$$

QR Decomposition

Assemble the matrix product

$$
\begin{array}{ccc}
\mathbf{A} = & \mathbf{Q} & \mathbf{R} \\
\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix} = &
\left[\frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \frac{1}{2\sqrt{15}} \begin{bmatrix} -4 \\ -3 \\ -2 \\ -1 \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \right] &
\begin{bmatrix} 3 & 15 \\ 0 & 2\sqrt{15} \end{bmatrix}
\end{array}
\tag{15}
$$

Least Squares Solution Using QR

Given $\mathbf{R}^{-1} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{2}\sqrt{\frac{5}{3}} \\ 0 & \frac{1}{2\sqrt{15}} \end{bmatrix}$,

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \mathbf{R}^{-1}\mathbf{Q}^* b = \frac{1}{360}\begin{bmatrix} 1733 \\ 3387 \end{bmatrix} \tag{16}$$

Exact agreement with (5).

Decomposition Factors

$$
\begin{array}{ccccc}
\mathbf{A} = \mathbf{U}\,\Sigma\,\mathbf{V}^* = & \left[\begin{array}{cc} \mathbf{U}_{\mathcal{R}(\mathbf{A})} & \mathbf{U}_{\mathcal{N}(\mathbf{A}^*)} \end{array}\right] & \left[\begin{array}{c} \mathbf{S} \\ \mathbf{0} \end{array}\right] & \left[\begin{array}{c} \mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} \end{array}\right] \\
 & \left[\begin{array}{cc} 9\times 2 & 9\times 7 \end{array}\right] & \left[\begin{array}{c} 2\times 2 \\ 7\times 2 \end{array}\right] & \left[\begin{array}{c} 2\times 2 \end{array}\right]
\end{array}
\tag{17}
$$

Singular Values

Find eigenvalues of $\mathbf{A}^*\mathbf{A}$. Compute characteristic polynomial:

$$\begin{aligned} p(\lambda) &= \det\left(\mathbf{A}^*\mathbf{A} - \lambda\mathbf{I}_2\right) \\ &= \lambda^2 - \mathrm{tr}\left(\mathbf{A}^*\mathbf{A}\right)\lambda + \det\left(\mathbf{A}^*\mathbf{A}\right) \end{aligned} \tag{18}$$

Find Roots of Characteristic Polynomial

Roots of Characteristic Polynomial:

$$\lambda_{\pm} = \frac{\text{tr}\left(\mathbf{A}^{*}\mathbf{A}\right) \pm \sqrt{\text{tr}^{2}\left(\mathbf{A}^{*}\mathbf{A}\right) - 4\text{tr}\left(\mathbf{A}^{*}\mathbf{A}\right)\det\left(\mathbf{A}^{*}\mathbf{A}\right)}}{2} \tag{19}$$

Trace and determinant in terms of mesh

$$\begin{aligned} \text{tr}\left(\mathbf{A}^{*}\mathbf{A}\right) &= \mathbf{1}^{*}\mathbf{1} + x^{*}x, \\ \det\left(\mathbf{A}^{*}\mathbf{A}\right) &= \left(\mathbf{1}^{\mathrm{T}}\mathbf{1}\right)\left(x^{*}x\right) - \left(\mathbf{1}^{\mathrm{T}}x\right)\left(x^{*}\mathbf{1}\right) \end{aligned} \tag{20}$$

Compute Singular Value Spectrum

$$\sigma\left(\mathbf{A}\right) = \sqrt{\lambda_{\pm}} = \sqrt{3\left(49 \pm \sqrt{2341}\right)} \tag{21}$$

Matrix of Singular Values

$$\mathbf{S} = \begin{bmatrix} \sigma_+ & 0 \\ 0 & \sigma_- \end{bmatrix} = \begin{bmatrix} \sqrt{3\left(49+\sqrt{2341}\right)} & 0 \\ 0 & \sqrt{3\left(49-\sqrt{2341}\right)} \end{bmatrix} \tag{22}$$

Sabot Matrix

Sabot matrix

$$\Sigma = \begin{bmatrix} \mathbf{S} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \sigma_+ & 0 \\ 0 & \sigma_- \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{23}$$

Matrix for the Solution Space

The domain matrix $\mathbf{V}$ is unitary and is therefore either a:

1. rotation matrix,
2. reflection matrix,
3. permutation matrix,

Matrix for the Solution Space

Guess the form of a rotation matrix $R_2(\theta)$:

$$\mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} = R_2(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{24}$$

Matrix for the Solution Space: Solving for θ

Guess the form of a rotation matrix $R_2(\theta)$:

$$\mathbf{A}^* \mathbf{A} = \mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} \, \mathbf{S}^2 \, \mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \mathbf{S}^2 \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \tag{25}$$

Matrix for the Solution Space: Solving for θ

Introducing δ:

$$\sigma_+^2 - \sigma_-^2 = \delta = 6\sqrt{2341} \tag{26}$$

Therefore

$$\begin{aligned} \mathbf{A}^*\mathbf{A} &= \mathbf{V}_{\mathcal{R}(\mathbf{A}^*)}\,\mathbf{S}^2\,\mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} \\ \left[\begin{array}{cc} \mathbf{1}^{\mathrm{T}}\mathbf{1} & \mathbf{1}^{\mathrm{T}}x \\ x^{\mathrm{T}}\mathbf{1} & x^{\mathrm{T}}x \end{array}\right] &= \left[\begin{array}{cc} \sigma_+^2\cos^2\theta + \sigma_-^2\sin^2\theta & \delta\cos\theta\sin\theta \\ \delta\cos\theta\sin\theta & \sigma_-^2\cos^2\theta + \sigma_+^2\sin^2\theta \end{array}\right] \end{aligned} \tag{27}$$

Matrix for the Solution Space: Solve for θ

We have multiple solution paths:

solution path	solution
$\mathbf{1}^{\mathrm{T}}\mathbf{1} = \sigma_+^2 \cos^2\theta + \sigma_-^2 \sin^2\theta$	$\cos\theta = \delta^{-\frac{1}{2}}\sqrt{\mathbf{1}^{\mathrm{T}}\mathbf{1} - \sigma_-^2}$
	$\sin\theta = \delta^{-\frac{1}{2}}\sqrt{\sigma_+^2 - \mathbf{1}^{\mathrm{T}}\mathbf{1}}$
$x^{\mathrm{T}}x = \sigma_-^2 \cos^2\theta + \sigma_+^2 \sin^2\theta$	$\cos\theta = \delta^{-\frac{1}{2}}\sqrt{x^{\mathrm{T}}x - \sigma_+^2}$
	$\sin\theta = \delta^{-\frac{1}{2}}\sqrt{x^{\mathrm{T}}x - \sigma_+^2}$

Relating Singular Values to the Mesh

$$
\begin{aligned}
\operatorname{tr}\left(\mathbf{A}^{*}\mathbf{A}\right) &= \operatorname{tr}\left(\mathbf{S}^{2}\right) \\
\mathbf{1}^{\mathrm{T}}\mathbf{1} + x^{\mathrm{T}}x &= \sigma_{+}^{2} + \sigma_{-}^{2}.
\end{aligned}
\tag{28}
$$

Establishing V Matrix

$$\mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \delta^{-\frac{1}{2}} \begin{bmatrix} \sqrt{\mathbf{1}^{\mathrm{T}}\mathbf{1} - \sigma_-^2} & -\sqrt{\sigma_+^2 - \mathbf{1}^{\mathrm{T}}\mathbf{1}} \\ \sqrt{\sigma_+^2 - \mathbf{1}^{\mathrm{T}}\mathbf{1}} & \sqrt{\mathbf{1}^{\mathrm{T}}\mathbf{1} - \sigma_-^2} \end{bmatrix}$$
$$= \delta^{-\frac{1}{2}} \begin{bmatrix} \sqrt{x^{\mathrm{T}}x - \sigma_+^2} & -\sqrt{\sigma_-^2 - x^{\mathrm{T}}x} \\ \sqrt{\sigma_-^2 - x^{\mathrm{T}}x} & \sqrt{x^{\mathrm{T}}x - \sigma_+^2} \end{bmatrix} \tag{29}$$

Matrix for the Data Space

$$
\begin{aligned}
\mathbf{U}_{\mathcal{R}(\mathbf{A})} &= \mathbf{A} \qquad \mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} \qquad \mathbf{S}^{-1} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \sigma_+^{-1} & 0 \\ 0 & \sigma_-^{-1} \end{bmatrix}
\end{aligned}
$$

Matrix for the Data Space

$$\mathbf{U}_{\mathcal{R}(\mathbf{A})} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} \sigma_+^{-1} \begin{bmatrix} \cos\theta - \sin\theta \\ \cos\theta - 2\sin\theta \\ \cos\theta - 3\sin\theta \\ \cos\theta - 4\sin\theta \\ \cos\theta - 5\sin\theta \\ \cos\theta - 6\sin\theta \\ \cos\theta - 7\sin\theta \\ \cos\theta - 8\sin\theta \\ \cos\theta - 9\sin\theta \end{bmatrix} & \sigma_-^{-1} \begin{bmatrix} \cos\theta + \sin\theta \\ 2\cos\theta + \sin\theta \\ 3\cos\theta + \sin\theta \\ 4\cos\theta + \sin\theta \\ 5\cos\theta + \sin\theta \\ 6\cos\theta + \sin\theta \\ 7\cos\theta + \sin\theta \\ 8\cos\theta + \sin\theta \\ 9\cos\theta + \sin\theta \end{bmatrix} \end{bmatrix} \tag{30}$$

Verify Decomposition

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix} = \mathbf{A} = \mathbf{U}_{\mathcal{R}(\mathbf{A})} \mathbf{S} \mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} \tag{31}$$

Error Propagation

$$\hat{\sigma}_0 = \sqrt{\frac{r^* r}{\mathbf{1}^{\mathrm{T}}\mathbf{1} - n}} \sqrt{\frac{1}{\sigma_+ \sigma_-}} \sqrt{\begin{bmatrix} \dfrac{\cos^2\theta}{\sigma_-^2} + \dfrac{\sin^2\theta}{\sigma_+^2} \\ \dfrac{\cos^2\theta}{\sigma_+^2} + \dfrac{\sin^2\theta}{\sigma_-^2} \end{bmatrix}},$$

$$\hat{\sigma}_1 = \sqrt{\frac{r^* r}{m - n}} \sqrt{\begin{bmatrix} \sigma_+^2 - \sqrt{\delta \mathbf{1}^{\mathrm{T}}\mathbf{1} - \sigma_-^2} \\ \sigma_-^2 + \sqrt{\delta x^{\mathrm{T}} x - \sigma_+^2} \end{bmatrix}}.$$

Hierarchy of Least Squares Methods

1. normal equations,
2. $\mathbf{QR}$ decomposition,
3. singular value decomposition,
4. iterative refinement

Why Use the Normal Equations?

The first method we are taught to solve least squares problems is the normal equations method. In the typical academic progression, the semester's preceding weeks have been spent developing elementary tools like Gaussian elimination and $\mathbf{LU}$ *decomposition for exact problems. The normal equations reduce inexact least squares problems to exact problems which can be solved with elementary methods. While an unwise choice for numerical problems, the normal equations are, as we shall see, an indispensable numerical tool.*

Deriving the Normal Equations

There are many ways to derive or posit the normal equations...

Deriving the Normal Equations

The normal equations arise from the minimization of t^2, the total squared error.

Deriving the Normal Equations: Definitions

Start with the linear system

$$\mathbf{A}\,x = b$$

where $\mathbf{A} \in \mathbb{C}^{m\times n}$, the data vector $b \in \mathbb{C}^m$, and the solution vector $x \in \mathbb{C}^n$. The residual error vector is

$$r(x) = \mathbf{A}\,x - b$$

Deriving the Normal Equations: Expand

Rewrite the total squared error to expose the differentiable terms containing x

$$\begin{aligned} t^2(x) &= (\mathbf{A}\,x - b)^*\,(\mathbf{A}\,x - b) \\ &= (\mathbf{A}\,x)^*\,\mathbf{A}\,x - (\mathbf{A}\,x)^*\,b - b^*\mathbf{A}\,x + b^*b \end{aligned}$$

Deriving the Normal Equations: Differentiate

The derivatives for $k = 1 \dots m$ are

$$\frac{d}{dx_k} r^2(x) = (\mathbf{A}\, x^*)\, \mathbf{A}\, e_{k,n} - b^* \mathbf{A}\, e_{k,n} = 0. \tag{32}$$

Deriving the Normal Equations: Differentiate

Compute the complex conjugate of both sides of this equation to find

$$((\mathbf{A}\,x^*)\,\mathbf{A}\,e_{k,n} - b^*\mathbf{A}\,e_{k,n})^* = e_{k,n}^{\mathrm{T}}\mathbf{A}^*\,(\mathbf{A}\,x - b) = 0. \qquad (33)$$

This establishes the fact that the kth row in $\mathbf{A}^*\mathbf{A}x$ is equal to the kth element in $\mathbf{A}^*b$, and so validates each row of the normal equations.

Resolving Data Space and Solution Space

Adjoint matrix erases the contribution of the null space component of the data vector

$$\mathbf{A}^* b = \mathbf{A}^* \left(b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)} \right) = \mathbf{A}^* b_{\mathcal{R}(\mathbf{A})} + \mathbf{0} = \mathbf{A}^* b_{\mathcal{R}(\mathbf{A})}. \tag{34}$$

Note the action of the matrix $\mathbf{A}$ itself

$$\mathbf{A}\, x = \mathbf{A} \left(x_p + x_h \right) = \mathbf{A}\, x_p + \mathbf{A}\, x_h = \mathbf{A}\, x_p + \mathbf{0} = \mathbf{A}\, x_p \tag{35}$$

Least Squares Problem Posed $\Rightarrow$ Solved

$$\begin{aligned} \mathbf{A}\,x = b \quad &\Longrightarrow \quad \mathbf{A}\left(x_p + x_h\right) = b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)} \\ &\Longrightarrow \quad \mathbf{A}\,x_p = b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)}. \end{aligned} \tag{36}$$

$$\begin{aligned} \mathbf{A}\,x_p = b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)} \quad &\Longrightarrow \quad \mathbf{A}^*\left(\mathbf{A}\,x_p\right) = \mathbf{A}^*\left(b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)}\right) \\ &\Longrightarrow \quad \mathbf{A}^*\mathbf{A}x_p = \mathbf{A}^* b_{\mathcal{R}(\mathbf{A})} \end{aligned} \tag{37}$$

Watch the Residual Error Vector

Residual error vector inhabits the left–hand null space, the null space of the column vectors:

$$\mathbf{A}\,x - b = r(x) \in \mathcal{N}(\mathbf{A}^{*}) \qquad \Longrightarrow \qquad r(x) \perp \mathbf{A}^{*}.$$

Therefore

$$\begin{array}{ccc} \mathbf{A}^{*} \quad r(x) & = & 0, \\ \Downarrow & & \\ \mathbf{A}^{*} \quad (\mathbf{A}\,x - b) & = & 0. \end{array}$$

Equivalence of Least Squares Solutions

The Least Squares solution is equivalent to the Normal Equations solution.

A Propitious Zero

Start with t^2 and stir in $\mathbf{0}$ in the guise:

$$\mathbf{A}x_N - \mathbf{A}x_N = \mathbf{0} \tag{38}$$

Thus relating both x_{LS} and x_N.

Connecting x_{LS} and x_N

$$\begin{aligned} t^2 &= \|\mathbf{A}\, x_{LS} - b\|_2^2 \\ &= \|\mathbf{A}\, x_{LS} + (\mathbf{A} x_N - \mathbf{A} x_N) - b\|_2^2 \\ &= \|\mathbf{A}\,(x_{LS} - x_N) + \mathbf{A} x_N - b\|_2^2 \,. \end{aligned} \tag{39}$$

Next up, Pythagoras.

Pythagoras

$$\begin{aligned} t^2 = &\left\| \mathbf{A}\left(x_{LS} - x_N\right)\right\|_2^2 + \left\|\mathbf{A}x_N - b\right\|_2^2 \\ &+ \left(\mathbf{A}\left(x_{LS} - x_N\right)\right)^* \left(\mathbf{A}x_N - b\right) + \left(\mathbf{A}x_N - b\right)^* \left(\mathbf{A}\left(x_{LS} - x_N\right)\right) \end{aligned} \tag{40}$$

The cross term will vanish...

Disappearing Cross Term

$$\left(\mathbf{A}\left(x_{LS}-x_{N}\right)\right)^{*}\left(\mathbf{A}x_{N}-b\right)=\left(x_{LS}-x_{N}\right)^{*}\mathbf{A}^{*}\left(\mathbf{A}x_{N}-b\right)=\mathbf{0}. \quad (41)$$

The cross term will vanish $(\mathcal{R}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A}^{*}))$...

Reduction of Terms

The least total squared error reduces to

$$t^2 = \left\| \mathbf{A} x_N - b \right\|_2^2 + \left\| \mathbf{A} \left(x_{LS} - x_N \right) \right\|_2^2 \tag{42}$$

Constraining the Norm

Right–hand side is the sum of two non-negative numbers which constrains $\|\mathbf{A}x_N - b\|_2^2$ to the extremal value of t^2, which implies an equivalence between x_{LS} and x_N. But this is not an equality, as, for example, x_N may have differing null space contributions. The desired equality follows from the second term

$$\|\mathbf{A}\left(x_{LS} - x_N\right)\|_2^2 = \mathbf{0} \qquad \Longrightarrow \qquad x_{LS} - x_N = \mathbf{0}. \tag{43}$$

Equivalence of Solutions

Conclusion:

$$x_{LS} = x_N \tag{44}$$

QR Decomposition

QR Decomposition is the default solution for the Least Squares problem

Least Squares Result

The Gram–Schmidt process is applied to the column vectors to orthogonalize the data space.

Singular Value Decomposition: Big Picture

Given a nonzero matrix $\mathbf{A} \in \mathbb{C}_{\rho}^{m \times n}$, a matrix with complex entries with m rows, n columns, and matrix rank $0 < \rho \leq \min(m, n)$, then there exists a decomposition of the form

$$\mathbf{A} = \mathbf{U}\,\Sigma\,\mathbf{V}^* \tag{45}$$

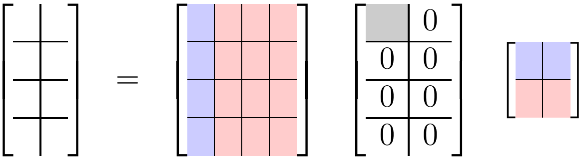
Singular Value Decomposition: Big Picture

1. column vectors of unitary matrix $\mathbf{V} \in \mathbb{C}^{n \times n}$ represent an orthonormal span of the row space (solution space),
2. column vectors of unitary matrix $\mathbf{U} \in \mathbb{C}^{m \times m}$ represent an orthonormal span of the column space (measurement space),
3. entries in the diagonal matrix $\mathbf{S} \in \mathbb{R}^{\rho \times \rho}$ contain the singular values; the square root of ordered, nonzero eigenvalues of the product matrix $\mathbf{A}^* \mathbf{A}$,
4. the sabot matrix $\Sigma \in \mathbb{R}^{m \times n}$ pads, $\mathbf{S}$, the matrix of singular values with zeros so that is conformable with both $\mathbf{U}$ and $\mathbf{V}$.

Conformability

$$\mathbf{A} \in \mathbb{C}_1^{4\times 2}$$

$$\mathbf{A} \quad = \quad \mathbf{U} \quad \Sigma \quad \mathbf{V}^*$$



(46)

$$m \times n \qquad m \times m \qquad \dim\left(\mathbf{A}\right) \qquad n \times n$$

Column Vectors Span Subspaces

$$\begin{aligned} \mathbf{V} &= \begin{bmatrix} v_1 & \dots & v_\rho & v_{\rho+1} & \dots & v_n \end{bmatrix}, \\ \mathbf{U} &= \begin{bmatrix} u_1 & \dots & u_\rho & u_{\rho+1} & \dots & u_m \end{bmatrix}. \end{aligned} \tag{47}$$

		range spaces			null spaces
$\mathcal{R}(\mathbf{A}^*)$	$=$	$\text{span}\{v_1, \dots, v_\rho\}$	$\mathcal{N}(\mathbf{A}^*)$	$=$	$\text{span}\{u_{\rho+1}, \dots, u_m\}$
$\mathcal{R}(\mathbf{A})$	$=$	$\text{span}\{u_1, \dots, u_\rho\}$	$\mathcal{N}(\mathbf{A})$	$=$	$\text{span}\{v_{\rho+1}, \dots, v_n\}$

Singular Value Decomposition as Column Vectors

$$
\begin{aligned}
\mathbf{A} &= \qquad\qquad\qquad \mathbf{U} \qquad\qquad\qquad\qquad \Sigma \qquad\qquad\quad \mathbf{V}^* \\
&= \left[\begin{array}{cccccc} u_1 & \dots & u_\rho & u_{\rho+1} & \dots & u_m \end{array}\right]
\left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \ddots & & \mathbf{0} \\ & & \sigma_\rho & \\ \hline & \mathbf{0} & & \mathbf{0} \end{array}\right]
\left[\begin{array}{c} {v_1}^* \\ \vdots \\ {v_\rho}^* \\ {v_{\rho+1}}^* \\ \vdots \\ {v_n}^* \end{array}\right]
\end{aligned}
\tag{48}
$$

Subspace Mappings: Watkins Diagram

$$
\begin{array}{ccccccc}
 & \mathbf{A} & & \quad & & \mathbf{A}^* & \\
v_1 & \xrightarrow{\sigma_1} & u_1 & & u_1 & \xrightarrow{\sigma_1} & v_1 \\
v_2 & \xrightarrow{\sigma_2} & u_2 & & u_2 & \xrightarrow{\sigma_2} & v_2 \\
\vdots & & \vdots & & \vdots & & \vdots \\
v_\rho & \xrightarrow{\sigma_\rho} & u_\rho & & u_\rho & \xrightarrow{\sigma_\rho} & v_\rho \\
v_{\rho+1} & \longrightarrow & \mathbf{0} & & u_{\rho+1} & \longrightarrow & \mathbf{0} \\
\vdots & & \vdots & & \vdots & & \vdots \\
v_m & \longrightarrow & \mathbf{0} & & u_n & \longrightarrow & \mathbf{0} \\
\hline
\mathbb{C}^m & \mapsto & \mathbb{C}^n & & \mathbb{C}^n & \mapsto & \mathbb{C}^m
\end{array}
$$

Minimize Total Squared Error

$$\begin{aligned} \|\mathbf{A}\,x - b\|_2^2 &= \|\mathbf{U}\,\Sigma\,\mathbf{V}^* - b\|_2^2 \\ &= \|\mathbf{S}\mathbf{V}^* b - \mathbf{U}^* b\|_2^2 \end{aligned} \tag{49}$$

Use Pythagorus to separate range and null space components.

Least Squares Result

$$\begin{aligned} \|\mathbf{A}\,x - b\|_2^2 &= \|\mathbf{U}\,\Sigma\,\mathbf{V}^* - b\|_2^2 \\ &= \underbrace{\left\|\mathbf{S}\,\mathbf{V}_{\mathcal{R}(\mathbf{A}^*)}x - \mathbf{U}^*_{\mathcal{R}(\mathbf{A})}b\right\|_2^2}_{\substack{\text{under control} \\ \text{adjust } x}} + \underbrace{\left\|\mathbf{U}^*_{\mathcal{N}(\mathbf{A}^*)}b\right\|_2^2}_{\text{no control}} \end{aligned} \tag{50}$$

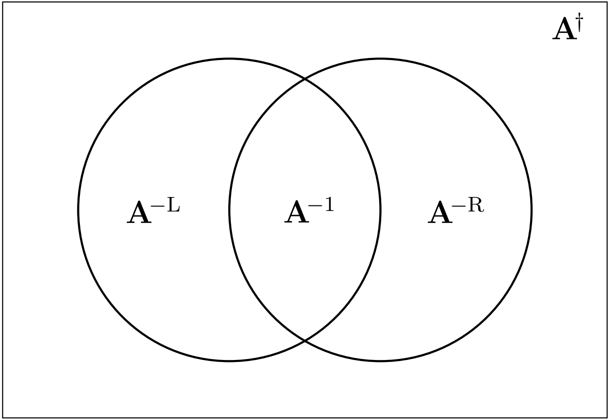
Minimize Total Squared Error

$$\mathbf{S}\mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} x_\star - \mathbf{U}^*_{\mathcal{R}(\mathbf{A})} b = \mathbf{0}. \tag{51}$$

Moore–Penrose Pseudoinverse Appears Magically

$$x_\star = x_p = \mathbf{V}_{\mathcal{R}(\mathbf{A}^*)}\mathbf{S}^{-1}\mathbf{U}^*_{\mathcal{R}(\mathbf{A})}b = \mathbf{A}^\dagger b. \tag{52}$$

Venn Diagram of Matrix Pseudoinverses



Moore–Penrose Pseudoinverse Decompositions

$$
\begin{array}{cccccc}
\text{type} & & \mathbf{U} & \Sigma & \mathbf{V}^{*} & \mathbf{A}^{\dagger} \\
\hline
① & \mathbf{A} = & \left[\begin{array}{c} \mathbf{U}_{\mathcal{R}(\mathbf{A})} \end{array}\right] & \left[\begin{array}{c} \mathbf{S} \end{array}\right] & \left[\begin{array}{c} \mathbf{V}^{*}_{\mathcal{R}(\mathbf{A}^{*})} \end{array}\right] & \mathbf{A}^{-1} \\
② & \mathbf{A} = & \left[\begin{array}{cc} \mathbf{U}_{\mathcal{R}(\mathbf{A})} & \mathbf{U}_{\mathcal{N}(\mathbf{A}^{*})} \end{array}\right] & \left[\begin{array}{c} \mathbf{S} \\ \mathbf{0} \end{array}\right] & \left[\begin{array}{c} \mathbf{V}^{*}_{\mathcal{R}(\mathbf{A}^{*})} \end{array}\right] & \mathbf{A}^{-\mathrm{L}} \\
③ & \mathbf{A} = & \left[\begin{array}{c} \mathbf{U}_{\mathcal{R}(\mathbf{A})} \end{array}\right] & \left[\begin{array}{cc} \mathbf{S} & \mathbf{0} \end{array}\right] & \left[\begin{array}{c} \mathbf{V}^{*}_{\mathcal{R}(\mathbf{A}^{*})} \\ \mathbf{V}^{*}_{\mathcal{N}(\mathbf{A})} \end{array}\right] & \mathbf{A}^{-\mathrm{R}} \\
④ & \mathbf{A} = & \left[\begin{array}{cc} \mathbf{U}_{\mathcal{R}(\mathbf{A})} & \mathbf{U}_{\mathcal{N}(\mathbf{A}^{*})} \end{array}\right] & \left[\begin{array}{cc} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right] & \left[\begin{array}{c} \mathbf{V}^{*}_{\mathcal{R}(\mathbf{A}^{*})} \\ \mathbf{V}^{*}_{\mathcal{N}(\mathbf{A})} \end{array}\right] & \mathbf{A}^{\dagger}
\end{array}
\tag{53}
$$

Least Squares Result

Table: Equivalent forms for the pseudoinverse matrix.

type	shape	row, col, rank	equivalence	form
①	square	$m = n = \rho$	$\mathbf{A}^{\dagger} = \mathbf{A}^{-1}$	
②	tall	$m > n,\ n = \rho$	$\mathbf{A}^{\dagger} = \mathbf{A}^{-\mathrm{L}}$	$= \left(\mathbf{A}^{*}\mathbf{A}\right)^{-1}\mathbf{A}^{*}$
③	wide	$m = \rho,\ n > m$	$\mathbf{A}^{\dagger} = \mathbf{A}^{-\mathrm{R}}$	$= \mathbf{A}^{*}\left(\mathbf{A}\mathbf{A}^{*}\right)^{-1}$
④	arbitrary	$m \ne n,$ $m \ne \rho,\ n \ne \rho$	$-$	

Pseudoinverse for a Vector

$$
\begin{aligned}
w^{\dagger} &= \mathbf{V} \qquad \Sigma^{(\dagger)} \qquad \mathbf{U}^{*} \\
&= \left[\ \mathbf{V}_{\mathcal{R}(\mathbf{A}^{*})}\ \right] \left[\ \mathbf{S}^{-1} \quad \mathbf{0}\ \right] \begin{bmatrix} \mathbf{U}^{*}_{\mathcal{R}(\mathbf{A})} \\ \mathbf{U}^{*}_{\mathcal{N}(\mathbf{A}^{*})} \end{bmatrix} \\
&= \left[\ 1\ \right] \left[\ \frac{1}{\sqrt{w^{*}w}} \quad 0 \quad \cdots \quad 0\ \right] \begin{bmatrix} \frac{1}{\sqrt{w^{*}w}} \begin{bmatrix} \overline{w}_1 \\ \vdots \\ \overline{w}_m \end{bmatrix} \\ \mathbf{U}^{*}_{\mathcal{N}(\mathbf{A}^{*})} \end{bmatrix} \\
&= \frac{w^{*}}{\|w\|_2^2},
\end{aligned}
\tag{54}
$$

Pseudoinverse for a Rank 1 Matrix

Given a rank 1 matrix, $\mathbf{A} \in \mathbb{C}_1^{m \times n}$, the pseudoinverse matrix is given by

$$\mathbf{A}^{\dagger} = \frac{\mathbf{A}^{*}}{\|\mathbf{A}\|_F^2} \tag{55}$$

Fundamental Theorem of Linear Algebra

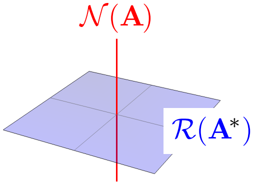
The Fundamental Theorem is an elegant statement which provides the stage upon which the drama of existence and uniqueness will unfold.

Hyperlinked references:

- Wolfram Mathworld
- Gilbert Strang
- The Four Fundamental Subspaces: 4 Lines
- Elementary Proofs

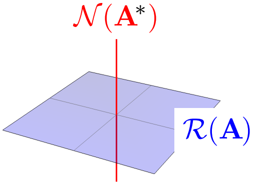
Fundamental Theorem of Linear Algebra

Row space	Mapping	Column space
Solution space		Measurement space
$n-$vectors		$m-$vectors



$$\mathbf{A}\colon \mathbb{C}^n \mapsto \mathcal{R}(\mathbf{A})$$

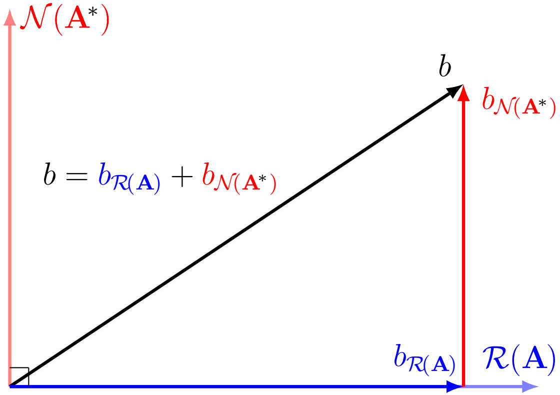
$$\mathcal{R}(\mathbf{A}^*) \leftarrow \mathbb{C}^m : \mathbf{A}^*$$



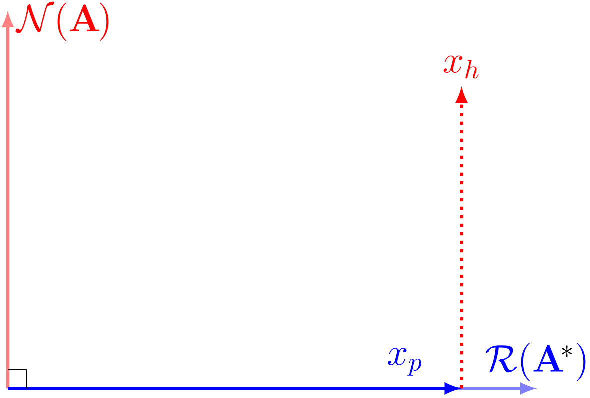
$$\mathbb{C}^n = \mathcal{R}(\mathbf{A}^*) \oplus \mathcal{N}(\mathbf{A})$$

$$\mathbb{C}^m = \mathcal{R}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^*)$$

Geometry of Least Squares Data Vector



Geometry of Least Squares Solution Vector



Least Squares Result

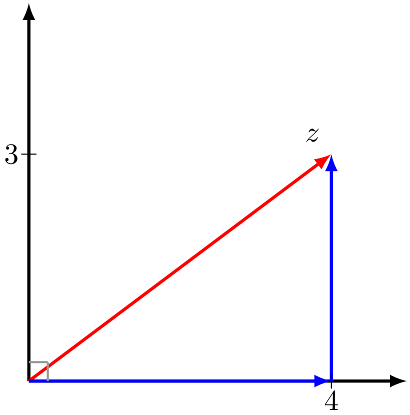
$$
\begin{aligned}
t^2 &= \left\| \mathbf{A}\, x_{LS} - b \right\|_2^2 && \text{minimization target} \\
&= \left\| \mathbf{A}(x_p + x_h) - \left(b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)} \right) \right\|_2^2 && \text{subspace decomposition} \\
&= \left\| \left(\mathbf{A}\, x_p - b_{\mathcal{R}(\mathbf{A})} \right) - b_{\mathcal{N}(\mathbf{A}^*)} \right\|_2^2 && \mathbf{A}\, x_h = \mathbf{0}, \text{ transitivity} \\
&= \left\| - b_{\mathcal{N}(\mathbf{A}^*)} \right\|_2^2 && \mathbf{A}\, x_p = b_{\mathcal{R}(\mathbf{A})} \\
&= {b_{\mathcal{N}(\mathbf{A}^*)}}^* b_{\mathcal{N}(\mathbf{A}^*)} && \text{definition of } 2-\text{norm} \\
&= \left\langle b_{\mathcal{N}(\mathbf{A}^*)}, b_{\mathcal{N}(\mathbf{A}^*)} \right\rangle && \text{definition of } 2-\text{norm}
\end{aligned} \tag{56}
$$

Different Notions of Length: $p-$norms

How far does a crow fly?
(Euclidean norm, Pythagorean norm, $2-$norm)

How far does a taxi drive?
(Manhattan norm, taxicab norm, $1-$norm)

Norms: Measuring Distance



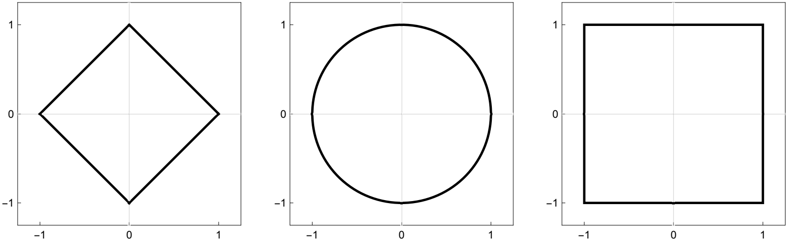
Different Notions of Length: $p-$norms

Common $p-$norms

$$
\begin{aligned}
\|p\|_1 &= |x| + |y| &= 7 \\
\|p\|_2 &= \sqrt{x^2 + y^2} &= 5 \\
\|p\|_\infty &= \max\left(|x|, |y|\right) &= 4
\end{aligned} \tag{57}
$$

Different Notions of Length: $p-$norms

$p = 1$	$p = 2$	$p = \infty$



$\|x + y\|_1 = 1$ $\|x + y\|_2 = 1$ $\|x + y\|_\infty = 1$

Norm Definitions

For $p \geq 1$ the $p-$norm for $x \in \mathbb{C}^m$

$$\|x\|_p = \left(\sum_{k=1}^{m} |x_k|^p \right)^{1/p} \tag{58}$$

Norm Definitions

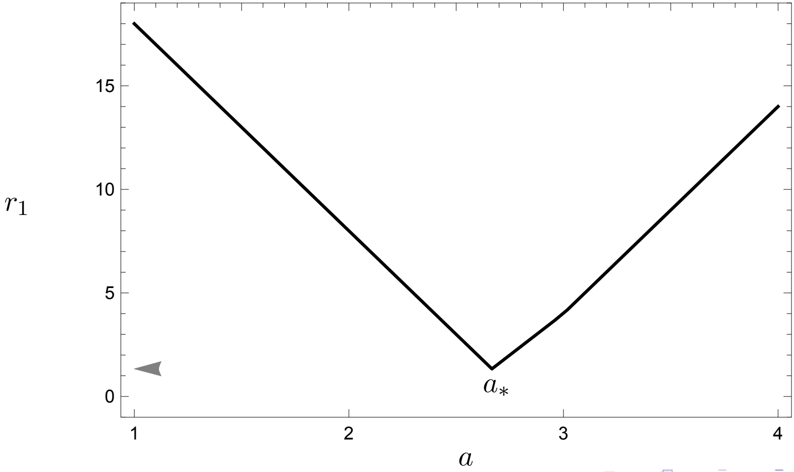
Properties of vector norms.

1. $\|x\|_p \geq 0$ with $\|x\|_p = 0 \iff x = \mathbf{0}$, (positive definite)
2. $\|\alpha x\|_p = |\alpha| \, \|x\|_p \, , \alpha \in \mathbb{C}$, (absolutely homogeneous)
3. $\|x + y\|_p \leq \|x\|_p + \|y\|_p$. (triangle inequality)

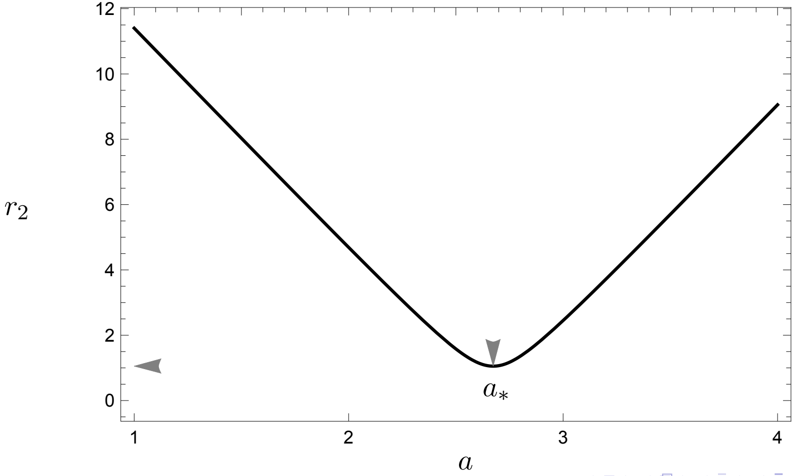
Why choose the $2-$norm

1. Gauss-Markov theorem
2. Differentiable
3. Precendence
4. Comfort

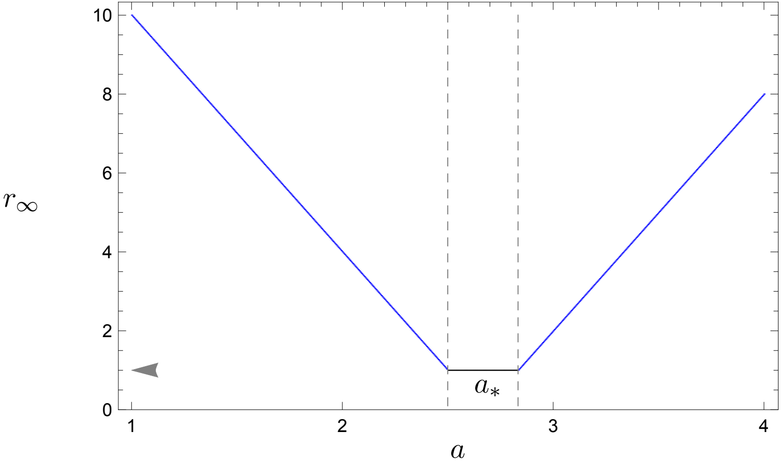
Differentiability: $1-$norm



Differentiability: $2-$norm



Differentiability: $\infty-$norm



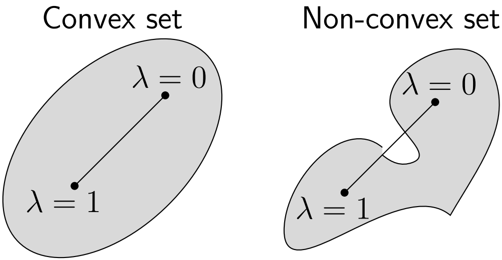
Convexity

Given a matrix $\mathbf{A} \in \mathbb{C}_{\rho}^{m \times n}$ with rank $(\rho \leq n)$, and a data vector $b \in \mathbb{C}^{m}$, then χ, the set of least squares minimizers,

$$\chi = \underset{x \in \mathbb{C}^{n}}{\arg\min} \left\| \mathbf{A}\, x - b \right\|_{2}\,, \tag{59}$$

is a convex set.

Seeing Convexity



Convexity

Consider two minimizers $x_1, x_2 \in \chi$ and a variation parameter $\lambda \in [0, 1]$. Establish that the linear combination is also in the set of minimizers:

$$\lambda x_1 + (1 - \lambda) x_2 \in \chi \tag{60}$$

Solve using matrix norm properties and the theorem of Pythagoras:

$$\begin{gathered} \|\mathbf{A}\left(\lambda x_1 + (1 - \lambda) x_2\right) - b\|_2^2 \leq \\ \lambda \|\mathbf{A}\, x_1 - b\|_2^2 + (1 - \lambda) \|\mathbf{A}\, x_2 - b\|_2^2 \end{gathered} \tag{61}$$

Convexity

Because the x variables are minimizers, the squared norms achieve the minimum value (least total squared error) t^2.

$$\lambda \left\| \mathbf{A}\, x_1 - b \right\|_2^2 + (1 - \lambda) \left\| \mathbf{A}\, x_2 - b \right\|_2^2 = \lambda t^2 + (1 - \lambda) t^2 = t^2 \tag{62}$$

Therefore, for all possible values of $\lambda \in [0, 1]$, the norm of the linear combination of the minimizers in (61) also maintains minimal value:

$$\left\| \mathbf{A} \left(\lambda x_1 + (1 - \lambda) x_2 - b \right) \right\|_2^2 = t^2. \tag{63}$$

Linear System

The design matrix $\mathbf{A} \in \mathbb{C}_{\rho}^{m\times n}$, the data vector $b \in \mathbb{C}^{m}$, and the solution vector $x \in \mathbb{C}^{n}$. If the equality holds, the solution is exact. Otherwise, we are finding an approximate solution.

$$\boxed{\mathbf{A}\,x = b} \tag{64}$$

Least Squares Problem

The definition of the least squares problem is a touchstone: x_{LS} is defined as the set of complex $n-$vectors such that the difference between prediction and measurement achieves minimum value in the $2-$norm

$$\boxed{x_{LS} = \underset{x \in \mathbb{C}^n}{\arg\min} \left\lVert \mathbf{A}\, x - b \right\rVert_2} \tag{65}$$

Decomposition of Least Squares Solution Set

Resolve the set of least squares minimizers x_{LS} into range space and null space components

$$\boxed{x_{LS} = x_{\mathcal{R}(\mathbf{A}^*)} + x_{\mathcal{N}(\mathbf{A})}} \tag{66}$$

Complete Least Squares Solution

The complete least squares solution is the typical combination of a vector $x_p \in \mathbb{C}^n$, a particular least squares solution (range space component), and a continuum solution, $x_h \in \mathbb{C}^n$, a homogeneous least squares solution (null space component).

$$\boxed{x_{LS} = x_p + x_h} \tag{67}$$

Least Squares Data Vector

Resolve the data vector b into a range space component $b_{\mathcal{R}(\mathbf{A})}$ and a null space component $b_{\mathcal{N}(\mathbf{A}^*)}$:

$$\boxed{b = b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)}} \tag{68}$$

Particular Least Squares Solution

Every least squares produces an exact solution and this exact solution is the particular least squares solution, x_p

$$\boxed{\mathbf{A}\, x_p = b_{\mathcal{R}(\mathbf{A})}} \tag{69}$$

Homogeneous Least Squares Solution

A linear system will present a non-trivial homogeneous least squares solution iff the right–hand null space is non-trivial. So if

$$\mathcal{N}(\mathbf{A}) \neq \{0\}, \tag{70}$$

there will be an infinite number of solutions signaled by the arbitrary vector y,

$$\boxed{x_h = (\mathbf{I}_n - \mathbf{A}^\dagger \mathbf{A}) y = \mathbf{P}_{\mathcal{N}(\mathbf{A})} y, \qquad y \in \mathbb{C}^n} \tag{71}$$

General Least Squares Solution

Usage is a matter of preference and context. Below the line, are reminders of the fundamental subspace decomposition for the solution.

$$\boxed{\begin{array}{cccccc} x_{LS} & = & \mathbf{A}^{\dagger}b & + & (\mathbf{I}_n - \mathbf{A}^{\dagger}\mathbf{A})y \\ & = & \mathbf{A}^{\dagger}b & + & \mathbf{P}_{\mathcal{N}(\mathbf{A})}y \\ \hline & = & x_p & + & x_h \\ & = & x_{\mathcal{R}(\mathbf{A}^*)} & + & x_{\mathcal{N}(\mathbf{A})} \end{array}} \tag{72}$$

Least Squares Residual Error Vector

The residual error vector resides in the left–hand null space $\mathcal{N}(\mathbf{A}^*)$. For an arbitrary vector $x \in \mathbb{C}^n$, the residual error vector is

$$\boxed{r(x) = \mathbf{A}\,x - b} \tag{73}$$

Least Squares Merit Function

The signature property of the method of least squares the minimization of the sum of the squares of the residual errors $r^*(x)r(x)$. The merit function is the target function for minimization:

$$\boxed{M(x) = r^*(x)r(x) = (\mathbf{A}x - b)^* (\mathbf{A}x - b)} \tag{74}$$

Least Total Squared Error

The method of least squares is the collection of tools and methods which minimizes the sum of the squares of the residual errors. When this minimal value is achieved, the magnitude of the merit function represents the least total square error, t^2. This value is achieved at x_{LS}:

$$\boxed{t^2 = r^*(x_{LS})r(x_{LS})} \tag{75}$$

Least Squares Uncertainty

For the overdetermined case when $m > n$, and with full column rank $(\rho = n)$, the vector quantifying least squares uncertainty is

$$\boxed{\sigma_k^2 = \frac{t^2}{m-n} \left(\mathbf{A}^* \mathbf{A}\right)^{-1}_{k,k}, \quad k = 1:n} \tag{76}$$

Be alert to context as the uncertainties are often called the errors in the fit parameters, or errors.

Least Squares Result

There are two styles to represent the the error terms: either as a separate vector or as part of the solution vectors:

$$\boxed{x_p = \begin{bmatrix} x_1 \\ \vdots \\ x_\rho \end{bmatrix} \pm \begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_\rho \end{bmatrix} \quad \text{or} \quad x_p = \begin{bmatrix} x_1 \pm \sigma_1 \\ \vdots \\ x_\rho \pm \sigma_\rho \end{bmatrix}} \tag{77}$$